

Amir Mahdian

FINAL YEAR COMPUTER SYSTEMS ENGINEERING STUDENT

+64 021 525 467 | amirmahdian16@gmail.com | [Personal Website](#) | [GitHub](#) | [LinkedIn](#) |  NZ Citizen

EDUCATION

The University of Auckland

Expected: Nov 2026

Bachelor of Engineering (Honours) – Computer Systems Engineering

- Relevant coursework: Software Quality, AI/ML & Deep Learning, Software Architecture, Advanced Software Engineering Methods, Digital Systems Design (FPGA/VHDL), OOP, Computer Networks, Embedded Systems

EXPERIENCE

NumberWorks'nWords

2022 – Present

Senior Mathematics Tutor | Auckland

- Tutored maths to students aged **5–16** across primary to high school curricula, managing up to **4 concurrent students** per session in a fast-paced environment over a **3+ year tenure** alongside full-time university study
- Mentored and onboarded junior tutors on effective teaching methodologies and session documentation standards across a student cohort of varying ages and abilities

PROJECTS

Honours Research Project — Robot Soccer System | UoA CARES Lab

- Building a multi-robot autonomous soccer system in Java/JavaFX with a computer vision perception pipeline (AprilTag localisation + ONNX-based ball detection) and a rule-based strategy engine for two-robot coordination
- Embedded firmware work on Arduino-based differential-drive robots with XBee wireless communication, PID motor control, and encoder feedback
- Stretch goal: reinforcement learning-based strategy in simulation

AI-Powered Network Intrusion Detection System

[GitHub Repository](#)

- Trained **Random Forest** and **neural network** classifiers on the **CICIDS-2017** dataset to detect **14 attack types** in real-time network traffic with a live threat dashboard deployed on **Streamlit Cloud**
- Built a **Scapy**-based feature extraction pipeline for live pcap traffic with attack classification and confidence scoring using **PyTorch** and **scikit-learn**

Real-Time Collaborative Code Editor

[GitHub Repository](#) | [Live Demo](#)

- Multi-user browser editor with real-time sync via **WebSockets** and **operational transforms**, **Monaco Editor**, shareable rooms, and coloured cursors using **Next.js**, **Node.js**, **Socket.io** and **Supabase**
- In-browser Python/JS execution via **WebAssembly (Pyodide)** and sandboxed **Web Workers** — no external API, no rate limits, zero-latency code execution

Encrypted P2P File Sharing CLI Tool

[GitHub Repository](#)

- Peer-to-peer file transfer in **Python** and **C** with manual **Diffie-Hellman key exchange**, **AES-256-CBC** encryption, **SHA-256** integrity checks, and live transfer progress — core protocol implemented **from scratch without libraries**

Personal Portfolio Website

[GitHub Repository](#) | [Live Demo](#)

- Animated portfolio built with **GSAP**, **ScrollTrigger**, custom cursor, and **Vanilla Tilt.js** hover effects; deployed on **GitHub Pages** with **async font loading**, **lazy images**, **WebP**, and **prefers-reduced-motion** accessibility support

LEADERSHIP & EXTRACURRICULAR

Albany United FC

2025

Assistant Coach — U15 Boys Championship | Auckland

- Co-led a **U15 squad** through a full championship season. Planned sessions, developed player skills, and coordinated match-day strategy; built **leadership** and **team communication** skills under competitive pressure

Albany United FC

2024 – Present

Striker — U23s Reserves | First Team | Auckland

- Compete in **U23 reserves** and First Team alongside full-time study and work; demonstrates consistent discipline and performance in a high-level team environment

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, VHDL, Assembly, MATLAB, HTML/CSS

Libraries/Frameworks: scikit-learn, PyTorch, Pandas, React, Next.js, Node.js, Express, Socket.io, FastAPI, Pyodide

Developer Tools: Git, GitHub, PostgreSQL, Supabase, Render, Vercel, Streamlit, GitHub Pages, GSAP, WebAssembly, JUnit, FPGA, Arduino, XBee, ONNX